EXPERIMENT 18

```
# Experiment: Perceptron Based Iris Classification
print("THASLIMA NASREEN J/192424152")
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay
# Load Iris dataset
iris = load_iris()
X = iris.data
y = iris.target
# Convert Iris classification into binary classification
# 0 = Setosa, 1 = Non-Setosa
y_binary = (y != 0).astype(int)
# Split dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y_binary,
    test_size=0.2,
    random_state=42,
    stratify=y_binary
)

# Standardize the features
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

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# Create Perceptron model
model = Perceptron(
    max_iter=1000,
    eta0=0.1,
    random_state=42
)
# Train the model
model.fit(X_train, y_train)
# Predict test data
y_pred = model.predict(X_test)
# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Actual Values:")
print(ytest)
print("\nPredicted Values:")
print(y_pred)
print("\nAccuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100, "%")
# Generate confusion matrix
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
# Display confusion matrix
disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=["Setosa", "Non-Setosa"]
)
disp.plot(cmap="Blues")
plt.title("Perceptron Based Iris Classification")
```

```
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.show()
```

OUTPUT:

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Actual Values:

[1 1 1 0 1 1 1 0 1 1 0 0 0 1 1 0 1 0 1 0 1 1 0 0 1 1 1 1 1 1]

Predicted Values:

[1 1 1 0 1 1 1 0 1 1 0 0 0 1 1 0 1 0 1 0 1 1 0 0 1 1 1 1 1 1]

Accuracy: 1.0

Accuracy Percentage: 100.0 %

Confusion Matrix:

[[10 0]

 [0 20]]